WORKED SOLUTION

STEP III 1997, Question 10

Mechanics • Rolling with Slipping

STEP III 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q10

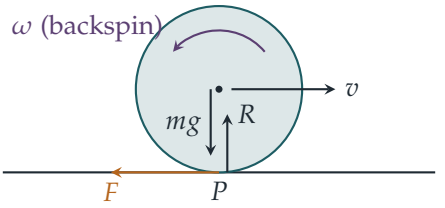
This question has a full guided treatment in *Beyond STEP*, Section 11.6, *Backspin*: https://beyondstep.keithmcnulty.org/ch11/backspin.html

The key idea

What matters is whether the bottom of the marble is sliding over the table. While it slides, the friction has a fixed size and a fixed direction, so both the linear and the angular motion are uniformly decelerated and we can write them down exactly. Sliding ends when the contact point comes to rest relative to the table, and after that the marble rolls with no friction at all. So the whole question is about which happens first: $v = 0$, $\omega = 0$, or the slip velocity reaching zero.

Solution

Model. The marble is a uniform solid sphere of mass m and radius a, so its moment of inertia about a diameter is $I = \frac{2}{5}ma^2$. Take v as the velocity of the centre (positive in the direction of v_0) and ω as the angular velocity in the *backward* sense (positive for backspin), so $v(0) = v_0 > 0$ and $\omega(0) = \omega_0 > 0$. The finger only starts the motion; afterwards the forces are the weight mg, the normal reaction $R = mg$ and the friction F at the contact point P.



With backspin, the bottom of the marble moves forward relative to the centre at speed

$a\omega$, so the velocity of the contact point is

$$s = v + a\omega.$$

Initially $s = v_0 + a\omega_0 > 0$: the marble skids forward over the table, so friction acts *backwards*, with constant magnitude F. Write $F = mk$.

Equations of motion while sliding. Resolving horizontally, $m\dot{v} = -F$. Taking moments about the centre, the backward friction at the bottom turns the marble in the forward-rolling sense, so it reduces the backspin:

$$\tfrac{2}{5}ma^2\,\dot{\omega} = -Fa \quad\Longrightarrow\quad \dot{\omega} = -\frac{5k}{2a}.$$

Both are constant, so as long as the marble keeps sliding forward,

$$v = v_0 - kt, \qquad a\omega = a\omega_0 - \tfrac{5}{2}kt, \qquad s = v_0 + a\omega_0 - \tfrac{7}{2}kt.$$

The slip velocity therefore falls to zero at

$$t_1 = \frac{2(v_0 + a\omega_0)}{7k},$$

and until then friction keeps the same direction, so these formulae hold on the whole interval $0 \le t \le t_1$, even if v or ω changes sign along the way.

Ask yourself What happens at $t = t_1$, and why doesn't the marble then start to slide the other way?

At t_1 the contact point is at rest, so $v = -a\omega$: the marble is rolling. If it carries on rolling at constant speed, nothing needs to push it, so friction (which only has to be as large as necessary once there is no slip) is zero, and rolling at constant velocity is consistent with all the equations. So from t_1 onwards the marble rolls uniformly with velocity

$$V = v(t_1) = v_0 - \tfrac{2}{7}(v_0 + a\omega_0) = \frac{5v_0 - 2a\omega_0}{7}.$$

Part (i)

To end up completely still, the final rolling velocity must be zero: $V = 0$, so $5v_0 = 2a\omega_0$, that is

$$\frac{v_0}{a\omega_0} = \frac{2}{5}.$$

Check that the picture makes sense: with this ratio, $v = 0$ at $t = v_0/k$ and $\omega = 0$ at $t = 2a\omega_0/(5k) = v_0/k$, the same moment, and this is also t_1. So linear and angular motion die away together while the marble is still sliding, and then it stays at rest (no slip, no friction).

Part (ii)

Now the marble must end up rolling backwards with speed $v_0/7$, so $V = -v_0/7$:

$$\frac{5v_0 - 2a\omega_0}{7} = -\frac{v_0}{7} \quad \Longrightarrow \quad 6v_0 = 2a\omega_0 \quad \Longrightarrow \quad \frac{v_0}{a\omega_0} = \frac{1}{3}.$$

With this ratio, v reaches zero at $t = v_0/k$, which is before $t_1 = 2(v_0 + 3v_0)/(7k) = 8v_0/(7k)$. At that instant the marble is momentarily stationary but still has backspin $\omega = \omega_0 - \frac{5v_0}{2a} = \frac{v_0}{2a} > 0$, so the bottom is still skidding forward and friction keeps pulling it back. At t_1 we get $v = -v_0/7$ and $a\omega = v_0/7$, so it rolls back towards its starting point at speed $v_0/7$, as required.

Ask yourself Is there a quicker way to get V without solving for t_1?

Yes. Take angular momentum about a fixed point O on the table's surface on the line of motion. The friction acts along the table surface, so it has no moment about O, and the moments of R and mg cancel. In the forward-rolling sense, the angular momentum about O is mva (the centre is at height a) minus the spin part $\frac{2}{5}ma^2\omega$, so this is conserved:

$$mv_0a - \tfrac{2}{5}ma^2\omega_0 = maV + \tfrac{2}{5}ma^2 \cdot \frac{V}{a} = \tfrac{7}{5}maV,$$

giving $V = (5v_0 - 2a\omega_0)/7$ again.

Checks

- With no spin ($\omega_0 = 0$) the formula gives $V = \frac{5}{7}v_0$, the standard result for a ball launched skidding without spin.
- With $v_0 = 0$ and backspin alone, $V = -\frac{2}{7}a\omega_0 < 0$: the marble moves backwards, as anyone who has flicked a backspun marble knows.
- Stepping the equations numerically (with the friction switching off once the slip velocity reaches zero) gives a final velocity of $-0.142857\ldots = -\frac{1}{7}$ for $v_0 = 1$, $a\omega_0 = 3$, and a dead stop for $v_0 = 2$, $a\omega_0 = 5$.

Common slips

- Stopping the clock when $v = 0$. The marble only stops sliding when $v + a\omega = 0$; at $v = 0$ it may still have spin, as in part (ii).
- Getting the sign of the torque wrong: backward friction at the bottom *reduces* backspin.
- Keeping the friction on after rolling starts. Once there is no slip, constant velocity needs no friction, so the friction drops to zero.

Beyond STEP

The friction here is Coulomb's model of sliding friction: a constant size while surfaces slip, and whatever is needed (up to a limit) when they don't. The angular-momentum shortcut is an instance of a general principle in rigid-body dynamics: choose the reference point so the awkward unknown force has no moment. Snooker and billiards players exploit exactly this physics with screw-back shots. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and rolling with slipping in this spirit.

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